

**VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.06)**

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1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator V1.06. Version 1.06 adds an isotope photon source library, activity unit conversion, line-by-line source calculation, source flux summation, and warning behavior for buildup requests outside the valid G-P mean free path range.

The baseline attenuation model, NIST XCOM interpolation, point source inverse-square flux model, and Geometric Progression (G-P) buildup calculation framework remain unchanged from the previously validated V1.05 implementation. Therefore, this report focuses only on the new V1.06 source library and source calculation features.

The V1.06 implementation supports selected major photon lines for the following isotope sources:

- Cs-137
- Co-60
- Am-241
- Ba-133
- Na-22
- Mn-54
- Co-57
- Zn-65
- Eu-152
- Ir-192

The source library stores photon energies in MeV and emission intensities as photons emitted per decay. Activity is converted internally to becquerels so each photon line emission rate can be calculated as activity multiplied by emission intensity.

V1.06 also improves robustness by preserving narrow-beam uncollided flux results when G-P buildup is requested outside the supported range. Instead of crashing or extrapolating buildup factors beyond the model range, the simulator skips invalid buildup for that photon line, reports a warning, and leaves the total buildup-corrected source flux unavailable.

2. METHODOLOGY AND PHYSICS MODELS

2.1 Previously Validated Attenuation and Buildup Models

The primary narrow-beam attenuation model remains unchanged. Shield transmission is still calculated using the Beer-Lambert Law, with NIST XCOM mass attenuation coefficients converted to linear attenuation coefficients using material density. The uncollided detector flux is still calculated using isotropic point-source geometric spreading.

The optional G-P buildup correction also remains based on the V1.05 implementation. V1.06 does not introduce a new buildup factor equation, new G-P coefficient tables, or a multilayer buildup treatment. The only new buildup-related behavior is source line warning handling when a requested buildup calculation exceeds the supported mean free path range.

2.2 Source Library Model

A source is represented as either a manual monoenergetic photon source or an isotope source. Isotope sources contain one or more PhotonLine objects:

PhotonLine = photon energy in MeV + emission intensity in photons per decay

2.3 Activity and Photon Emission Rate

For isotope source calculations, activity is converted to becquerels. Since 1 Bq = 1 decay/s, the photon emission rate for a given photon line is calculated as:

$$S_i = A \cdot I_i$$

where:

- S_i is the photon emission rate for the i -th photon line, in photons/s.
- A is the source activity, in Bq.
- I_i is the emission intensity of the i -th photon line, in photons/decay.

2.4 Multi-Line Source Summation

The simulator performs one shielding calculation for each photon line in an isotope source. Each line receives its own attenuation coefficient interpolation, transmission calculation, mean free path calculation, uncollided flux result, and optional buildup correction.

The total uncollided flux is calculated by summing the uncollided flux from all photon lines:

$$\phi_{uncollided, total} = \sum_{i=1}^N \phi_{uncollided, i}$$

If buildup is requested and successfully applied to every photon line, the total buildup-corrected source flux is calculated by summing the line-by-line buildup-corrected flux values.

$$\phi_{buildup, total} = \sum_{i=1}^N \phi_{buildup, i}$$

If buildup is skipped for any photon line, the total buildup-corrected source flux is not reported because the total would otherwise combine corrected and uncorrected line results.

2.5 Buildup Range Handling

V1.06 enforces the G-P buildup mean free path range by rejecting buildup calculations above 40 MFP. For source calculations, this rejection is handled as a non-fatal warning. The simulator still

reports narrow-beam uncollided flux because that calculation remains mathematically defined, but it does not extrapolate the G-P buildup factor outside the supported range.

3. TEST CASE RESULTS

3.1 Activity Unit Conversion Tests

The validation runner checks the activity conversions used by the isotope source model.

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	1 Ci	$3.7 \cdot 10^{10} Bq$	$3.7 \cdot 10^{10} Bq$	PASS
2	1 mCi	$3.7 \cdot 10^7 Bq$	$3.7 \cdot 10^7 Bq$	PASS
3	1 μ Ci	$3.7 \cdot 10^4 Bq$	$3.7 \cdot 10^4 Bq$	PASS

3.2 Isotope Source Library Tests

The validation runner verifies that the source library contains ten isotope keys and that each source returns the expected number of selected photon lines.

Test Case	Input Parameters	Expected Result	Actual Result	Status
4	Source Library	10 isotope keys	10 isotope keys	PASS
5	Cs-137	1 photon line	1 photon line	PASS
6	Co-60	2 photon lines	2 photon lines	PASS
7	Am-241	1 photon line	1 photon line	PASS
8	Ba-133	7 photon lines	7 photon lines	PASS
9	Na-22	2 photon lines	2 photon lines	PASS
10	Mn-54	1 photon line	1 photon line	PASS
11	Co-57	2 photon lines	2 photon lines	PASS
12	Zn-65	2 photon lines	2 photon lines	PASS
13	Eu-152	9 photon lines	9 photon lines	PASS
14	Ir-192	10 photon lines	10 photon lines	PASS

The validation runner also confirms that every photon line has energy greater than zero and intensity greater than zero. This verifies that source library entries are physically usable by the shielding calculation functions.

3.3 Isotope Source Calculation Tests

Each isotope source was run through the narrow-beam calculator using a 1 Ci source, 5 cm of lead, and a 100 cm detector distance. These tests validate source bookkeeping rather than validating the previously validated attenuation equations.

Test Case	Input Parameters	Expected Result	Actual Result	Status
15	All isotope sources	Line result count equals source photon line count.	Line result count equals source photon line count.	PASS
16	All isotope sources	Photon rates equal activity times intensity.	Photon rates equal activity times intensity.	PASS
17	All isotope sources	Total uncollided flux equals sum of line fluxes.	Total uncollided flux equals sum of line fluxes.	PASS

This confirms that multi line isotope sources are processed as independent photon line calculations and then summed correctly at the source result level.

3.4 Manual Source Regression Test

The validation runner checks that manual monoenergetic source mode still reproduces the V1.05 lead shielding regression case. This confirms that V1.06 source wrapping did not alter the validated single-energy calculation pathway.

Test Case	Input Parameters	Expected Result	Actual Result	Status
18	0.6617 MeV, $S = 3.7 \cdot 10^{10}$ photons/s, 5 cm lead, $r = 100$ cm	$\phi_{\text{uncollided}} \approx 5.40662 \cdot 10^2$ photons/cm ² /s	$\phi_{\text{uncollided}} \approx 5.40662 \cdot 10^2$ photons/cm ² /s	PASS
19	0.6617 MeV, $S = 3.7 \cdot 10^{10}$ photons/s, 5 cm lead, $r = 100$ cm, buildup enabled	$\phi_{\text{buildup}} \approx 1.42612 \cdot 10^3$ photons/cm ² /s	$\phi_{\text{buildup}} \approx 1.42612 \cdot 10^3$ photons/cm ² /s	PASS

3.5 Isotope Source Example Tests

Manual command-line tests were also performed for Cs-137 and Co-60 using 1 Ci activity, 5 cm of lead, 100 cm detector distance, and buildup enabled.

Test Case	Input Parameters	Expected Result	Actual Result	Status
20	Cs-137, 1 Ci, 5 cm lead, $r = 100$ cm, buildup enabled	Single 0.661657 MeV line produces valid	$\phi_{\text{uncollided}} \approx 4.598798 \cdot 10^2$ photons/cm ² /s $\phi_{\text{buildup}} \approx$	PASS

		uncollided and buildup-corrected flux.	$1.213016 \cdot 10^3$ photons/cm ² /s	
21	Co-60, 1 Ci, 5 cm lead, r = 100 cm, buildup enabled	Two photon lines are calculated and summed.	$\phi_{\text{uncollided total}} \approx 2.079821 \cdot 10^4$ photons/cm ² /s $\phi_{\text{buildup total}} \approx 5.611395 \cdot 10^4$ photons/cm ² /s	PASS

The Cs-137 isotope result is lower than the manual V1.05 regression case because the isotope source uses the physical photon emission intensity of 0.851 photons/decay rather than assuming one photon emitted per decay.

3.6 Invalid Buildup Range Warning Tests

V1.06 adds warning behavior for buildup requests outside the supported 40 MFP range. The validation runner and manual tests confirm that narrow-beam flux is still reported while buildup-corrected flux is withheld.

Test Case	Input Parameters	Expected Result	Actual Result	Status
22	Am-241, 1 Ci, 5 cm lead, r = 100 cm, buildup enabled	No total buildup-corrected flux; warning generated	Total MFP = 290.34, $\phi_{\text{uncollided}} \approx 8.530218 \cdot 10^{-122}$ photons/cm ² /s, warning generated	PASS
23	Manual Source 0.059541 MeV, S = $3.7 \cdot 10^{10}$ photons/s, 5 cm lead, r = 100 cm, buildup enabled	No total buildup-corrected flux; warning generated	$\phi_{\text{uncollided total}} \approx 2.376105 \cdot 10^{-121}$ photons/cm ² /s, warning generated	PASS

This behavior is physically appropriate because the uncollided transmission remains mathematically defined, but the G-P buildup factor should not be extrapolated to 290 mean free paths.

3.7 Error Handling Tests

Test Case	Input Parameters	Expected Result	Actual Result	Status
24	Unsupported activity unit: banana	ValueError	ValueError	PASS
25	Unsupported isotope source: fake_isotope	ValueError	ValueError	PASS
26	Negative activity: -1 Ci	ValueError	ValueError	PASS

4. ASSUMPTIONS AND LIMITATIONS

1. Single Layer Homogeneous Buildup Only: G-P buildup correction remains supported only for single-layer homogeneous shielding cases.
2. No Multilayer Buildup Treatment: Multilayer buildup is not implemented. The simulator rejects buildup mode for multiple shielding layers because applying a single material buildup factor to mixed materials may produce misleading results.
3. Supported Buildup Materials: V1.06 supports G-P exposure buildup correction for lead, aluminum, ordinary concrete, water, tungsten, copper, and tin.
4. Unsupported Buildup Materials: Barite concrete and polyethylene remain unsupported for buildup correction in V1.06, although they remain available for narrow-beam attenuation calculations.
5. Point-Source Geometry: The source is still modeled as an isotropic point source.
6. Narrow-Beam Transmission Still Used as Baseline: The buildup-corrected flux is calculated by multiplying the narrow-beam uncollided flux by the G-P buildup factor.
7. Coefficient Interpolation: G-P coefficients are linearly interpolated between tabulated energies. This is separate from the NIST XCOM attenuation interpolation, which uses log-log interpolation.
8. Exposure Buildup Coefficients: The implemented coefficients are exposure buildup coefficients. Future versions may distinguish exposure, dose, and energy buildup factors if additional coefficient tables are added.
9. Simple 1D Geometry: The simulator assumes shielding thickness lies along the source to detector path. Therefore, total shielding thickness cannot exceed detector distance.
10. Selected Photon Lines: The V1.06 isotope library uses selected major photon lines, not complete emitted decay spectra.
11. Emitted Photons Before Source Self-Attenuation: Photon intensities are treated as emitted photons per decay before source encapsulation, source self-attenuation, air attenuation, or detector-window effects.
12. No Dose Conversion: V1.06 reports photon flux, not dose rate, exposure rate, air kerma, or effective dose.

13. G-P Range Limit: Buildup correction is only applied through 40 mean free paths. If a photon line exceeds this range, narrow-beam flux is reported but buildup is skipped with a warning.

5. CONCLUSION

The Shielding Attenuation Simulator V1.06 successfully adds isotope source modeling and validates the new source-level calculation pathway without changing the previously validated attenuation and buildup physics models.

Validation confirmed that activity unit conversions are correct, the ten-source isotope library is structurally consistent, photon-line rates are calculated from activity and intensity, total uncollided flux equals the sum of line-by-line flux values, and Co-60 multi-line buildup totals equal the sum of valid line-by-line buildup-corrected fluxes.

The V1.05 manual source regression case remains preserved, confirming that V1.06 did not break the original monoenergetic calculation pathway. V1.06 also improves robustness by warning when buildup is requested outside the valid G-P mean-free-path range while still reporting narrow-beam uncollided flux.

V1.06 represents a practical improvement over V1.05 by allowing shielding calculations to be driven by real isotope activities and selected photon emission lines. Future work should focus on target classes, minimum-thickness calculations, dose-rate conversion, and eventually comparison against Monte Carlo transport tools.

References

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